SUMMATIVE ASSIGNMENT

For this project, you are provided with five datasets:

- Option A: Life Expectancy Data — *Life_expectancy.csv*
- Option B: Sleep Efficiency Data — *Sleep_Efficiency.csv*
- Option C: Traffic Accidents Data — *Traffic_Accidents.csv*
- Option D: Students Mental Health Data – *Students_Mental_Health.csv*
- Option E: World Development Indicators – *WDI_data.csv*

Your group should select **ONE** dataset to investigate. The aim of the project is to demonstrate your ability to apply analytical techniques learned in the module, formulate meaningful questions, and draw insights from real-world data with minimal guidance. You are expected to think creatively and critically about what the dataset allows you to explore.

You will submit a written report addressing the following components:

1. Develop a set of economically or socially relevant questions that can be answered using your chosen dataset. Clearly state each question and explain why it is interesting or important. Your questions should guide the direction of your analysis.

2. Discuss whether your dataset requires cleaning or preprocessing. This may include handling missing values, correcting data types, removing inconsistencies, or addressing outliers. Explain any steps you take and justify them.

3. Use appropriate exploratory data analysis (EDA) techniques and visualisation tools to provide an overview of the dataset. This may include summary statistics, distributions, correlations, trends, or initial patterns you observe.

4. Apply suitable analytical methods to address the questions you formulated in Part 1. Support your findings with clear visualisations and well-reasoned interpretations. You are encouraged to go beyond basic summaries where appropriate and demonstrate initiative in your choice of methods.

Overall word limit: 3000 words

SUBMISSION INSTRUCTIONS

Your completed assignment must be uploaded to Ultra no later than 12:00 midday on 15th January

The report for the assignment should be submitted in PDF format. Your submission should also include corresponding .ipynb and .xlsx files.

A penalty will be applied for work uploaded after 12:00 midday as detailed in the Economics SharePoint site. You must leave sufficient time to fully complete the upload process before the deadline and check that you have received a receipt. At peak periods, it can take up to 30 minutes for a receipt to be generated.

Assignments should be typed, using 1.5 spacing and an easy-to-read 12-point font. Assignments and dissertations must not exceed the word count indicated in the module handbook/assessment brief.

The word count should:

- *Include* all the text, including title, preface, introduction, in-text citations, quotations, footnotes and any other items not specifically excluded below.
- *Exclude* diagrams, tables (including tables/lists of contents and figures), equations, executive summary/abstract, acknowledgements, declaration, bibliography/list of references and appendices. However, it is not appropriate to use diagrams or tables merely as a way of circumventing the word limit. If a student uses a table or figure as a means of presenting his/her own words, then this is included in the word count.

Examiners will stop reading once the word limit has been reached, and work beyond this point will not be assessed. Where a student has intentionally misrepresented their word count, the Department may treat this as an offence under Section IV of the General Regulations of the University. Extreme cases may be viewed as dishonest practice under Section IV, 5 (a) (x) of the General Regulations.

Very occasionally it may be appropriate to present, in an appendix, material which does not properly belong in the main body of the assessment but which some students wish to provide for the sake of completeness. Any appendices will not have a role in the assessment - examiners are under no obligation to read appendices and they do not form part of the word count. Material that students wish to be assessed should always be included in the main body of the text.

Guidance on referencing can be found on Durham University website and in the Economics SharePoint site.

MARKING CRITERIA AND FEEDBACK

We use **Grade Descriptors** and **Marking Criteria** defined by the type of assessment to evaluate your work.

Rather than providing numerical scores for individual criteria, markers will decide a mark based on an overall, holistic assessment that reflects the work's quality in relation to the grade descriptors.
Please refer to the Postgraduate Grade Descriptors and the Marking Criteria for Written Report.

Individual and generic feedback will be provided via Blackboard (see PGT Assessment and Feedback Dates).

GENERATIVE AI

The use of Generative AI is allowed. See the Department's Policy and Guidance on the use of Generative AI for more details.

PLAGIARISM AND COLLUSION

Students suspected of plagiarism, either of published work or the work of other students, or of collusion will be dealt with according to School and University guidelines.

END OF ASSESSMENT